AI-Powered Candidate Ranking System

Team Information

Team Name: Tech Roots

Team Members:

- Devanath N (Leader)
- Thanusri Daivala
- Avishi Pandey
- Naveen Raj

Problem Statement

Develop an AI-powered candidate ranking system to identify and rank the most suitable candidates from a large candidate dataset using data-driven evaluation techniques.

Project Overview

Recruitment platforms often receive thousands of candidate applications, making manual screening time-consuming and inefficient. The objective of this project is to automate the candidate evaluation process by analyzing candidate profiles and generating a ranked list of the most suitable candidates.

The proposed solution processes candidate data, evaluates experience, education, and technical skills, calculates candidate scores, and generates ranked recommendations for recruitment purposes.

Proposed Solution

The system automatically analyzes candidate profiles and ranks candidates based on predefined scoring criteria.

Key features include:

- Candidate data loading
- Feature engineering
- Candidate scoring
- Candidate ranking

- Submission generation
- Output validation

This approach reduces manual effort and provides a consistent and scalable candidate evaluation process.

Candidate Evaluation Approach

The system evaluates candidates using the following signals:

- Technical Skills
- Years of Experience
- Education Tier
- Candidate Profile Information
- Professional Background

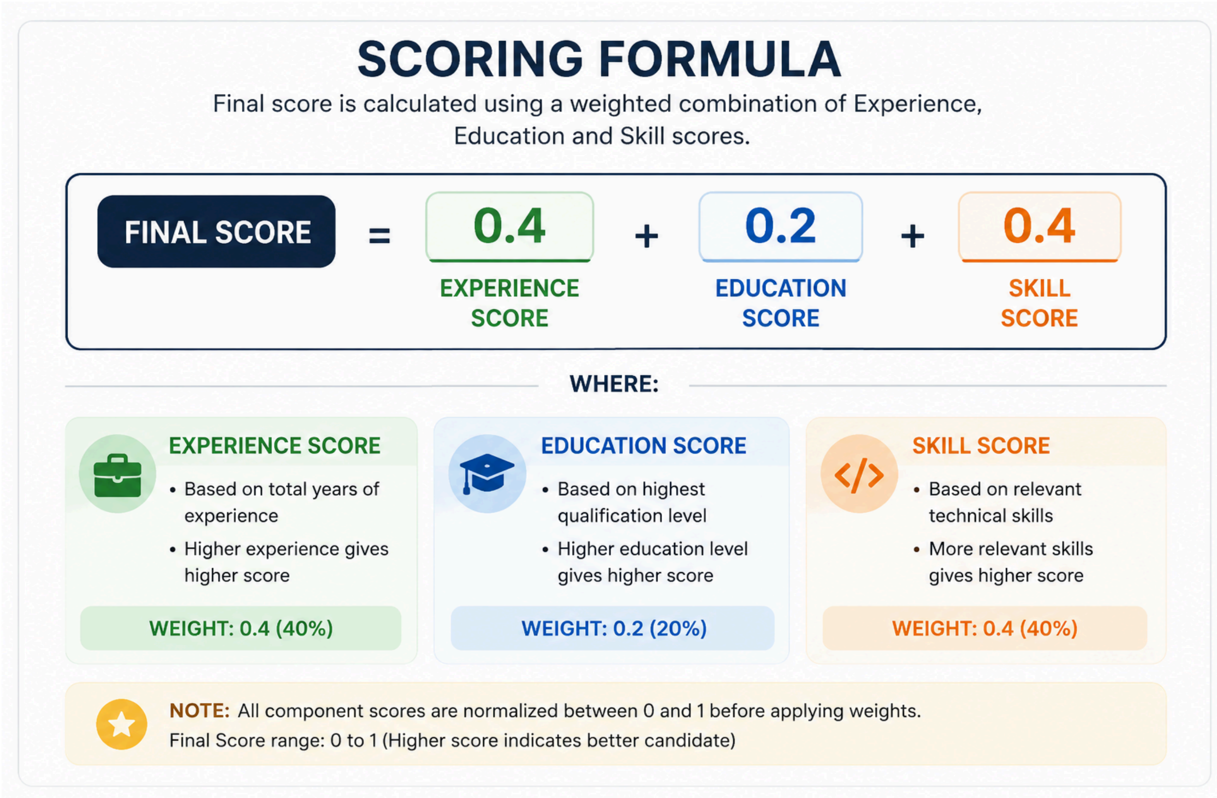
Unlike traditional keyword-based matching systems, this solution uses feature engineering and weighted scoring to evaluate multiple candidate attributes together and generate more meaningful rankings.

Ranking Methodology

The ranking process consists of:

1. Loading candidate profiles
2. Extracting candidate features
3. Calculating individual scores
4. Combining scores using weighted scoring
5. Ranking candidates based on final scores

Scoring Formula



Explainability and Data Validation

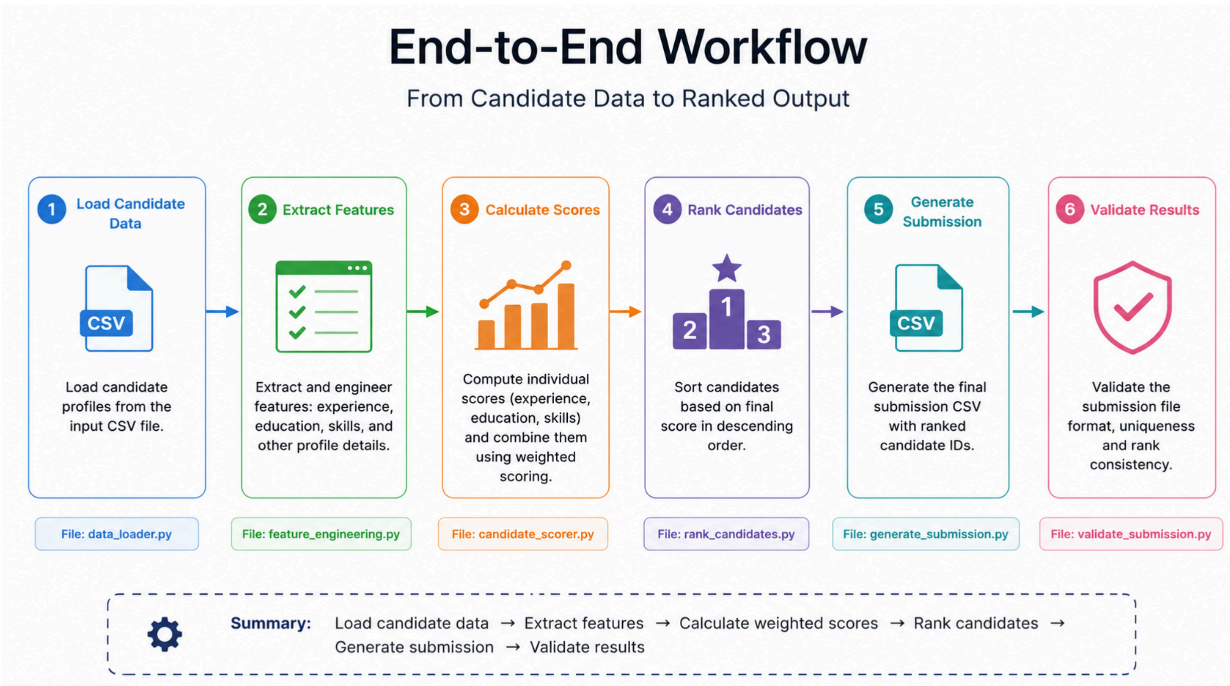
Ranking decisions are based entirely on candidate profile information.

The system:

- Uses only available candidate data
- Applies predefined scoring rules
- Avoids unsupported assumptions
- Produces transparent and explainable rankings

Profiles with weaker experience, education, or skill indicators naturally receive lower rankings.

End-to-End Workflow



System Architecture

The solution is composed of the following modules:

Data Loader

Loads candidate profiles from the dataset.

Feature Engineering

Extracts experience, education, and skill-based features.

Candidate Scoring Engine

Calculates weighted scores for each candidate.

Ranking System

Sorts candidates based on final scores.

Submission Generator

Creates the final submission file.

Validation Module

Verifies submission correctness and formatting.

Results and Performance

Achievements

- Processed 100,000 candidate profiles
- Successfully generated ranked candidate recommendations
- Produced a validated submission file
- Completed the end-to-end ranking workflow

Performance

The solution uses lightweight feature engineering and weighted scoring techniques, allowing efficient execution without requiring expensive computational resources.

Technologies Used

- Python
- JSON
- CSV
- VS Code
- PowerShell
- Git
- GitHub

These technologies were selected because they provide efficient support for data processing, candidate evaluation, ranking, and project management.

Project Progress

Phase 1 – Project Setup ✅

Phase 2 – Data Loading ✅

Phase 3 – Schema Analysis ✅

Phase 4 – Feature Engineering ✅

Phase 5 – Scoring Engine ✅

Phase 6 – Ranking System ✅

Phase 7 – Submission Generator ✅

Phase 8 – Validation ✅

Overall Progress: 100%

Submission Assets

Project Title:
AI-Powered Candidate Ranking System

GitHub Repository:
https://github.com/Deva-1706/AI-Powered-Candidate-Ranking-System

Presentation Deck:
AI_Powered_Candidate_Ranking_System.pdf

Ranked Output File:
DEVANATH.csv

Validation Status:
Submission Successfully Validated

Technologies Used:
Python, JSON, CSV, Git, GitHub, VS Code

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Team Leader:
Devanath N

Conclusion

The AI-Powered Candidate Ranking System successfully automates the candidate evaluation and ranking process. By combining feature engineering, weighted scoring, and automated ranking techniques, the system efficiently identifies suitable candidates from large datasets. The project demonstrates practical applications of data processing, analytics, and recruitment automation while producing validated outputs suitable for large-scale candidate screening.